VOUCH: Anytime-Valid Certificates for Machine Unlearning via Paired-Canary Hypothesis-Betting

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Abstract

Machine unlearning methods for large language models are deployed to honour deletion requests, yet they ship with no statistical certificate that forgetting actually occurred. The evaluations in common use are fixed-sample tests: they are unsound as membership evidence unless what enters training is randomised, their error guarantees collapse the moment an auditor looks at the evidence as it accumulates, and they lean on a retrained reference model that is unaffordable at scale and, for multi-stage pipelines, not even well defined. We introduce *VOUCH*, an anytime-valid statistical certification framework that removes all three obstacles at once. At fine-tuning time the provider plants *paired ghost canaries*: exchangeable twin records of which a fair coin admits one into training (and later into the forget set, so that it traverses the entire unlearning pipeline) while the other is never shown to any model. Because exact unlearning leaves the two twins statistically interchangeable, the within-pair score difference is symmetric *by construction*; this hands the verifier an exact, distribution-free, finite-sample null that needs neither shadow models nor a retrained reference nor any assumption about the model. Over this null VOUCH runs betting e-processes that remain valid at every stopping time simultaneously, and from them derives three coupled objects: an anytime-valid confidence sequence on the residual membership advantage, an $(\boldsymbol{\varepsilon}, \boldsymbol{\alpha})$-forgetting certificate obtained by sequential two-sided equivalence testing, and a revocation alarm. We prove that valid composition runs in opposite directions for the two claims—certificates compose by consensus, alarms by product—and show empirically that the intuitive alternatives are invalid, falsely certifying up to **99.6%** of the time. Across a two-thousand-seed calibration study under adversarial optional stopping and end-to-end pipelines on the TOFU and MUSE-News benchmarks over several architectures, VOUCH certifies genuine unlearning, revokes un-unlearned models within a handful of canary pairs, and—by pairing every certificate with a utility reading—exposes gradient-ascent unlearning that forgets by destroying the model. Verification costs on the order of $\mathbf{10^4}$ forward passes, roughly two orders of magnitude below shadow-model evaluation.

Keywords: machine unlearning, anytime-valid inference, e-values, membership inference, certification, large language models

1 Introduction

When a user invokes the right to erasure, a model provider must do more than run an unlearning routine: it must be able to *show*, to a regulator or an internal audit function, that the routine worked. This obligation is now concrete. The EU AI Act's audit provisions, Article 17 of the GDPR, and a growing set of national data-protection regimes all presume that erasure can be demonstrated, not merely asserted. The algorithmic side of machine unlearning has risen to meet the demand: gradient ascent and its stabilised successors, negative preference optimisation [1, 2], representation misdirection [3], and task-vector negation now remove targeted knowledge from large models at modest cost. The *verification* side has not kept pace, and it is verification, not optimisation, that a regulator cares about.

Three obstacles stand between current practice and a defensible certificate, and it is worth stating them plainly because VOUCH is organised around dismantling each one. *First*, a high membership-inference score on a supposedly forgotten example is not evidence that the example was ever trained on. Distribution shift and the model's own priors confound the signal, and Zhang et al. [4] have shown formally that membership cannot be established without controlling the randomness of what enters training. *Second*, deletion requests arrive as a stream and auditors inspect the evidence as it accrues, stopping when they are satisfied; every fixed-sample test in current use—the forget-quality metric of Maini et al. [5], the leakage metric of Shi et al. [6], the shadow-model panels of Hayes et al. [7]—has its error guarantee voided by exactly this optional stopping, and we measure the damage in Section 5.1. *Third*, the comparators these tests rely on—a model retrained from scratch without the deleted data—are prohibitively expensive at the scale of modern language models and, as Yu et al. [8] show, are not even well defined for the multi-stage pipelines used in practice, because path-dependence means no unique retrained model exists to compare against.

VOUCH dissolves all three obstacles with a single design decision and a single inferential engine. The design decision is the *paired ghost canary*. At fine-tuning time we generate exchangeable twin records, and for each pair we flip a fair coin to decide which twin is admitted into training; the admitted twin is later appended to the forget set, so that it passes through the very same unlearning pipeline as organic data, while its twin is withheld from every model. This construction is what makes the audit sound: under exact unlearning the model is independent of the coin, so the two twins are exchangeable and the difference between their scores is symmetric about zero *regardless of the model, the scoring function, or the data distribution*. We therefore obtain an exact, distribution-free, finite-sample null that we generated ourselves, rather than one we must assume. The engine is the betting e-process from game-theoretic statistics, which turns this null into a test whose validity holds not at one predetermined sample size but at every stopping time at once, by Ville's inequality.

Streaming deletions and continuous public monitoring, fatal to a fixed-sample test, are precisely what an e-process is built to survive.

Building the framework and running it end-to-end forced four corrections and refinements on the initial design, and because each turned out to matter in practice we foreground them rather than bury them. We found that the natural way to combine several attack scores into one certificate—bet on an adaptively weighted mixture of their signs—is *anytime-invalid*, and in simulation it falsely certified a leaking model 99.6% of the time; the sound alternative requires every score's own process to concur. We found that certificates and alarms must compose in *opposite* directions across deletion waves, and that multiplying certificate e-values, the instinctive move, certifies histories that contain a genuinely bad wave. We found on real language models that gradient-ascent unlearning frequently *over-forgets*, driving the forgotten twins to scores conspicuously *below* their never-seen partners—membership leakage that a one-sided test happily certifies and that a two-sided test must catch. And we found, when we corrected the sample-size analysis against the information-theoretic lower bound, that an earlier back-of-envelope estimate understated the required cohort by threefold. Each of these is now a theorem or a measured result rather than a footnote.

The contributions of this paper are as follows. We formulate unlearning certification as sequential two-sided equivalence testing against a declared attack class, with an exact distribution-free null supplied by paired ghost canaries (Section 3). We give the supporting theory: exact validity at any stopping time, sound composition across scores and across deletion waves in both directions, power that matches the Kullback–Leibler information limit, a magnitude-aware revocation arm that is exact under the symmetry null, and a bridge from certified $(\varepsilon_u, \delta_u)$-unlearning (Section 4). Finally we report a comprehensive empirical study (Section 5): calibration under adversarial optional stopping over two thousand seeds; end-to-end certification of gradient ascent, GradDiff, negative preference optimisation, and retraining-from-scratch on the TOFU and MUSE-News benchmarks across two model architectures, with utility guardrails and dose–response calibration; recoverability probes that relearn, quantise, and jailbreak the unlearned model; a model axis spanning 2019 to 2026; and cost measurements showing that a certificate takes seventy-six seconds on a single mid-range GPU. All code, all results, and the fault-tolerant harness that produced them on free ephemeral compute are released.

2 Related work

Our work sits at the meeting point of three literatures—unlearning algorithms, the verification of unlearning, and game-theoretic statistics—and it is the gap between the first two that game-theoretic statistics turns out to close. We survey each in turn and then state precisely how VOUCH differs from its nearest neighbours.

2.1 Unlearning algorithms and their benchmarks

Exact unlearning by architectural design, exemplified by sharded training [9], retrains only the shards touched by a deletion; it is sound but cannot be applied to a model that was not trained for it, which is the situation every deployed language model is in. The practical response has therefore been *approximate* unlearning, and for large language models a compact toolbox has emerged: gradient ascent on the forget set and its retain-regularised variant GradDiff; negative preference optimisation [1], which replaces the unbounded ascent objective with a bounded preference loss and which its authors introduced precisely to avoid the catastrophic collapse that ascent invites; the simplified reappraisal SimNPO [2]; representation misdirection [3]; and task-vector negation. These methods are evaluated on purpose-built benchmarks—TOFU's fictitious-author question answering [5], MUSE's six-way news-and-books evaluation [6], and the hazardous-knowledge multiple-choice of WMDP [3]—and we treat all of them, methods and benchmarks alike, as *subjects* that VOUCH certifies rather than as competitors.

What this literature has increasingly documented, however, is that the metrics used to declare success are unreliable. A growing body of adversarial work shows that apparent forgetting is often mere suppression: Łucki et al. [10] recover ostensibly unlearned capabilities through lightweight attacks; Zhang et al. [11] show that simply quantising an unlearned model to four bits resurrects the forgotten knowledge; Deeb and Roger [12] recover most of the pre-unlearning accuracy by fine-tuning on adjacent facts; and Fan et al. [13] formalise relearning attacks and connect robustness to sharpness-aware minimisation. The evaluations themselves have come under equally sharp scrutiny. Feng et al. [14] demonstrate that current LLM-unlearning evaluations can be made to both overstate and understate success; Wei et al. [15] show that facts presumed forgotten persist through correlated knowledge; Xu et al. [16] distinguish genuine erasure from suppression at the representation level and find that behaviour is easily restored by minimal fine-tuning; and Bărbulescu and Triantafillou [17] document how per-sequence memorisation strength governs how hard a datum is to remove. The recurring message across all of this work is that *evaluation*, not optimisation, is the bottleneck—which is exactly the problem VOUCH addresses, and which motivates our recoverability probes (R-VOUCH) and our insistence on pairing every certificate with a utility reading.

2.2 Verifying and certifying unlearning

The earliest guarantees came from *certified removal* [18–21], which bounds the distance between the unlearned and retrained models in a differential- privacy-style $(\varepsilon_u, \delta_u)$ sense. These results are elegant but confined to convex or near-convex regimes, and their bounds become vacuous at the scale of modern language models; we use them not as a baseline but as a semantic anchor, showing in Proposition 6 that any $(\varepsilon_u, \delta_u)$-certified algorithm passes VOUCH. The verification of *arbitrary* unlearning has proven far harder. Thudi et al. [22] argue that approximate unlearning cannot be audited from weights alone and that auditable definitions are a prerequisite for any guarantee; Zhang et al. [23] show that a dishonest provider can defeat both existing families of verification strategies while passing them; and, most consequentially for our design, Zhang et al. [4] establish that membership-inference scores *cannot* prove that a model was trained on given data unless the inclusion of that data was randomised—the soundness principle that our fair-coin canary design is built to satisfy. Two further impossibility results bound what any behavioural certificate can claim: Yu et al. [8]

show that path-dependence makes retrain equivalence unattainable for multi-stage pipelines, which is why VOUCH's null is internally generated rather than referenced to a retrained model, and the reversibility findings above imply that no behavioural method can certify parameter-space erasure—a limit we state explicitly rather than paper over.

A handful of recent efforts move toward statistical certification, and these are our closest neighbours. Pandey et al. [24] give a hypothesis-testing notion of certified unlearning, but it is fixed-sample and tied to a Gaussian high-dimensional regression regime rather than to LLM-scale behavioural auditing. Pawelczyk et al. [25] use data-poisoning as a lens on unlearning failure but provide no error-controlled certificate. Cryptographic proof-of-unlearning [26] attests that the correct computation *ran*, which is orthogonal to whether forgetting *holds* statistically and with which VOUCH composes. The recent survey of Xue et al. [27] organises this fast-moving area and confirms that no existing method delivers what VOUCH does: an anytime-valid, distribution-free certificate of bounded residual influence. The broader programme is surveyed by Liu et al. [28].

2.3 Game-theoretic statistics

The inferential engine we borrow is that of safe, anytime-valid inference: e-values, test martingales, and betting confidence sequences [29–33]. The defining property, traceable to Ville's inequality [34], is that a nonnegative martingale started at one exceeds $1/\alpha$ with probability at most α *at every stopping time simultaneously*; this is precisely what a streaming, continuously-monitored deletion audit requires, and precisely what a fixed-sample test cannot offer. The betting perspective [32, 35] additionally supplies confidence sequences whose bets adapt online to near-optimal growth, which we use both for the certificate and for its quantitative confidence-sequence face. The one adjacent application of these tools is to privacy auditing: Steinke et al. [36] audit differential privacy in a single training run, and González et al. [37] and Csillag and Mesquita [38] bring sequential e-value machinery to DP auditing. The distinction is fundamental to the hypothesis structure. Privacy auditing *lower*-bounds the privacy loss of a *training* algorithm by mounting a successful attack; VOUCH *upper*-bounds the residual influence *after unlearning* by showing that no attack in a declared class succeeds—an equivalence test, not a significance test, run over ghost twins that have passed through the unlearning pipeline. To our knowledge, VOUCH is the first application of game-theoretic statistics to unlearning certification.

2.4 Positioning

The contrast with our nearest neighbours can be summarised sharply. Fixed-sample paired or two-sample tests—including the KS-based forget-quality metric of Maini et al. [5] and the min-k% leakage metric of Shi et al. [6]—are invalidated by optional stopping and typically require a retrained reference; the sound-membership results of Zhang et al. [4] tell us *why* controlled randomness is indispensable but stop at a fixed-n membership proof rather than a sequential forgetting certificate; shadow-model evaluators [7] are descriptive and cost dozens of fine-tunes; certified removal [18, 19] does not scale; and DP auditing [36, 37] solves the opposite, lower-bounding, problem. VOUCH is, to our knowledge, the first framework to formulate unlearning certification as sequential equivalence testing and to deliver exact, finite-sample, anytime-valid certificates of bounded residual influence at LLM scale, with soundness inherited from a randomised paired-canary design rather than from distributional assumptions or shadow models.

3 Setting and the VOUCH protocol

This section develops the protocol in full: the actors and threat model, the canary construction that manufactures the null, the certified quantity, the betting engine with its explicit wealth dynamics, and finally the assembled protocol as pseudocode (Algorithm 1) and as a flowchart (Figure 1).

3.1 Actors, pipeline, and threat model

The provider fine-tunes a base model on $D_{\text{keep}} \cup D_{\text{forget}} \cup C$, where C is the canary set described below. When deletion requests arrive, an unlearning algorithm $\mathcal{U}$—negative preference optimisation, GradDiff, and so on—produces an unlearned model M_u from the fine-tuned model, using the forget set augmented with the relevant canaries. A verifier, holding only black-box query access to M_u together with a canary manifest whose hash was published before unlearning began, runs VOUCH and either issues, withholds, or revokes a certificate. Throughout we assume the honest-but-verifiable setting: the provider genuinely executes $\mathcal{U}$ and the open question is whether it worked and how well, under quantified error control. A provider willing to special-case the committed canaries is outside our scope and, as prior work establishes, can only be addressed by cryptographic attestation, with which VOUCH composes.

3.2 Paired ghost canaries

A templated generator emits exchangeable twin pairs (c_i^0, c_i^1) for $i = 1, \ldots, m$. The two twins in a pair are independent draws from the same conditional template distribution—the same surface form populated with independently sampled fictitious entities carrying secrets of at least forty bits of entropy—so that within a pair they are exchangeable by construction. We match the template to the host corpus, using question–answer pairs for TOFU and record-style sentences for news text, so that the canaries do not stand out. For each pair an independent fair coin $b_i \sim \text{Bern}(\frac{1}{2})$ selects the *in-twin* $c_i^{b_i}$, which we insert into the fine-tuning corpus with a repetition factor $r_i \in \{1, 2, 4, 8\}$ (these strata give us a dose–response curve) and later append to the forget set, so that it traverses the unlearning pipeline exactly as organic forget data does. The *ghost twin* $c_i^{1-b_i}$ is never shown to any model. Before unlearning begins the provider publishes the commitment

$$\text{commit} \;=\; H\big(\{(c_i^0,\, c_i^1,\, b_i,\, r_i)\}_{i=1}^m\big), \qquad H = \text{SHA-256}, \tag{1}$$

which binds the pairs and the coins and prevents the audit from being gamed after the fact: at verification time the manifest is revealed and checked against (1) before a single query is made.

3.3 Scores and the certified quantity

We fix a declared class $\mathcal{F}$ of scalar score functions—the token-normalised negative log-likelihood of the secret span, the min-k% token log-probability, the base-model-calibrated loss ratio, and optionally a linear probe on the hidden state—each aggregated over Q fixed query prompts. For pair i and score s we form the within-pair difference and its sign,

$$D_i^{(s)} = s(M_u, c_i^{\text{in}}) - s(M_u, c_i^{\text{ghost}}), \qquad Z_i^{(s)} = \mathbf{1}\{D_i^{(s)} > 0\}, \tag{2}$$

breaking ties by a committed fair coin. Writing $p^{(s)} = \Pr[Z^{(s)} = 1]$ and $\Delta^{(s)} = 2p^{(s)} - 1$, the quantity we certify is $\sup_{s\in\mathcal{F}} |\Delta^{(s)}|$, the largest residual advantage any score in the class can extract. We are deliberate that this is influence extractable *through* $\mathcal{F}$ on the canary population: it is the strongest target compatible with the impossibility results for behavioural audits, and we make no claim about erasure in parameter space. The certificate itself is worth pinning down precisely, since everything that follows is engineered to earn it.

Definition 1 ((ε,α)-forgetting certificate) Fix a tolerance $\varepsilon \in (0,1)$, an error budget $\alpha \in (0,1)$, and the boundary $p_0 = \frac{1}{2} + \frac{\varepsilon}{2}$. An (ε,α)-forgetting certificate for M_u over the class $\mathcal{F}$ is a data-dependent stopping decision that asserts $\sup_{s\in\mathcal{F}} |\Delta^{(s)}| < \varepsilon$, issued by a procedure with the guarantee that, whenever in truth $\sup_{s\in\mathcal{F}} |\Delta^{(s)}| \geq \varepsilon$, the probability that a certificate is *ever* issued—at any stopping time, under any monitoring rule—is at most α.

The two-sidedness in Definition 1 is deliberate: $\Delta^{(s)} \leq -\varepsilon$, the over-forgetting regime in which the in-twin scores conspicuously *below* its never-seen partner, is membership leakage exactly as much as $\Delta^{(s)} \geq \varepsilon$ is, and Section 5 shows that gradient-ascent unlearning lands there in practice.

3.4 The betting engine

All three of VOUCH's outputs are built from one primitive, the *betting e-process*, and it is worth writing its dynamics out explicitly because the validity proofs in Section 4 reduce to inspecting them. A gambler starts with unit wealth and, before pair i is revealed, stakes a predictable fraction λ_i (chosen from the information $\mathcal{F}_{i-1}$ revealed so far) on the sign Z_i deviating from a hypothesised mean. Against the null $H_0^+ : \mathbb{E}[Z] \geq p_0$ ("the in-twins still score high") and its mirror $H_0^- : \mathbb{E}[Z] \leq 1 - p_0$, the wealth processes for score s are

$$E_t^{s,+} = \prod_{i\leq t}\bigl(1 + \lambda_i^{s,+}\,(p_0 - Z_i^{(s)})\bigr), \qquad E_t^{s,-} = \prod_{i\leq t}\bigl(1 + \lambda_i^{s,-}\,(Z_i^{(s)} - (1 - p_0))\bigr), \tag{3}$$

with stakes confined to $\lambda_i^{s,\pm} \in [0,\, 1/(1-p_0))$ so that wealth can never go negative. Under any distribution in the respective null each factor has conditional mean at most one, so each process is a nonnegative supermartingale with initial value one, and Ville's inequality converts wealth into evidence: the chance that a supermartingale ever exceeds $1/\alpha$ is at most α. Wealth above $1/\alpha$ is therefore a licence to reject that null at whatever time we happen to be looking—the anytime validity on which the whole framework rests.

How the stakes are chosen determines power, not validity, which frees us to be aggressive. VOUCH plays a uniform mixture of eight experts and lets their wealth arbitrate: six constant-fraction bettors $\lambda = c \cdot \lambda_{\max}$ with $c \in \{0.02, 0.05, 0.1, 0.2, 0.4, 0.7\}$, an online-Newton-step (ONS) learner, and a truncated Krichevsky–Trofimov (KT) plug-in. The ONS expert follows the coin-betting recursion

$$g_i = \frac{p_0 - Z_i}{1 + \lambda_i(p_0 - Z_i)}, \qquad A_i = A_{i-1} + g_i^2, \qquad \lambda_{i+1} = \Pi_{[0,\lambda_{\max}]}\Big(\lambda_i + \tfrac{\eta\, g_i}{A_i}\Big), \tag{4}$$

with $A_0 = 1$, $\eta = \frac{1}{2}$, and Π the clip onto the admissible range; its log-wealth regret against the best constant bet in hindsight is $O(\log t)$. The KT expert bypasses stake selection altogether: it estimates the mean by $q_i = (\sum_{j<i} Z_j + \frac{1}{2})/i$, truncates q_i to the alternative side of p_0, and multiplies wealth by the plug-in likelihood ratio

$$e_i \;=\; Z_i\,\frac{q_i}{p_0} \;+\; (1-Z_i)\,\frac{1-q_i}{1-p_0}, \tag{5}$$

which has conditional mean at most one under the null (it is linear in the mean, equals one at p_0, and decreases into the null) and achieves the optimal growth rate up to a $\frac{1}{2t}\log t$ redundancy. Because a convex combination of e-processes with a fixed prior is again an e-process, the mixture inherits exact validity while paying only a $\log 8$ additive cost in log-wealth relative to its best expert—power adapts online with nothing to tune.

The same wealth idea, run over a grid rather than at a single boundary, yields the quantitative face of the certificate. For every candidate mean m on a grid of 1001 points in $[0,1]$ we maintain two wealth processes betting that the truth is above and below m, with empirical-Bernstein (aGRAPA) stakes

$$\lambda_i(m) \;=\; \Pi_{[0,\,\lambda_{\max}(m)]}\Big(\tfrac{\hat\mu_{i-1} - m}{\hat\sigma_{i-1}^2 + (\hat\mu_{i-1} - m)^2}\Big), \tag{6}$$

where $\hat\mu_{i-1}$ and $\hat\sigma_{i-1}^2$ are the running mean and variance of the signs. The set of grid values whose wealth on both sides stays below $2/\alpha$, intersected over time,

$$C_t \;=\; \bigcap_{u \le t} \big\{\, m:\; \max\big(W_u^{\uparrow}(m),\, W_u^{\downarrow}(m)\big) < 2/\alpha \,\big\}, \tag{7}$$

is a $(1-\alpha)$ confidence sequence for $p^{(s)}$: it covers the truth at every time simultaneously, so the auditor may glance at it whenever they please. Mapped through $\Delta = 2p - 1$ it reports the residual advantage with its anytime-valid uncertainty.

The revocation arm inverts the role of the null. Under *exact* unlearning, Theorem 1 makes every $Z_i^{(s)}$ an exact fair coin, so here—and only here—adaptive score weighting is sound. We run, per score and per direction, e-processes against $\mathbb{E}[Z] = \frac{1}{2}$ and combine them with predictable wealth-proportional weights: with $w_{s,i} \propto \exp(\log E_{i-1}^s)$ computed before pair i is revealed,

$$R_t = \prod_{i \leq t} \sum_{s \in \mathcal{F}} w_{s,i} \frac{E_i^s}{E_{i-1}^s}, \tag{8}$$

which is an e-process because each factor is a predictable convex combination of conditional-mean-one increments. When $R_t \geq 1/\alpha$ the verifier holds anytime-valid evidence of residual memorisation and the certificate is withheld or revoked; the weights meanwhile steer wealth toward whichever score discriminates best, which is exactly what an alarm should do.

3.5 The protocol end to end

Algorithm 1 assembles the pieces into the three-phase protocol, and Figure 1 shows the same pipeline as a flowchart. Phase 0 runs at design time: generate the cohort, flip the coins, publish the commitment, fine-tune with the in-twins planted. Phase 1 is the provider's unlearning step, in which the in-twins ride along with the organic forget set. Phase 2 is the audit: reveal and check the manifest, then stream the pairs in random order, updating the certificate processes, the confidence sequences, and the revocation mixture after each pair, and act on whichever boundary is crossed first. R-VOUCH then re-runs the same loop on probed variants of M_u—after relearning on public data, after 4-bit quantisation, and under adversarial prompt wrappers—and the robust certificate requires every probe to pass. Streaming deletion waves each draw a fresh cohort with independent coins; Theorem 4 dictates how their verdicts may be combined.

4 Theory

Our guarantees rest on one structural fact—the null we plant is exact—and on the supermartingale property of betting. The proofs are short, and their brevity is the point: the design does the work, so we give every proof in full. Throughout, $\mathcal{F}_{i-1}$ denotes the verifier's filtration, generated by everything revealed through pair $i-1$; all bets and mixture weights are $\mathcal{F}_{i-1}$-measurable (predictable).

Theorem 1 (Exact, distribution-free null) *Suppose the unlearned model M_u is conditionally independent of the coin b_i given the pair (c_i^0, c_i^1); exact unlearning and retraining-from-scratch both satisfy this. Then within-pair exchangeability gives $D_i^{(s)} \overset{d}{=} -D_i^{(s)}$, so $Z_i^{(s)} \sim \mathrm{Bern}(\frac{1}{2})$ exactly—for any model, any score, and any set of query prompts.*

Proof Condition on the unordered pair $\{c_i^0, c_i^1\}$ and the past $\mathcal{F}_{i-1}$. The twins are exchangeable within the pair by construction, and under the hypothesis M_u is conditionally

Algorithm 1 The VOUCH protocol

Phase 0 (design time, before fine-tuning)

generate twin pairs $\{(c_i^0, c_i^1)\}_{i=1}^m$; draw coins $b_i \sim \mathrm{Bern}(\frac{1}{2})$ and strata $r_i \in \{1, 2, 4, 8\}$

$\textsc{In} \leftarrow \{c_i^{b_i}\}$; $\textsc{Ghost} \leftarrow \{c_i^{1-b_i}\}$ ▷ ghosts are never shown to any model

publish commit $= H(\{(c_i^0, c_i^1, b_i, r_i)\})$ ▷ SHA-256, Eq. (1)

fine-tune M_{ft} on $D_{\mathrm{keep}} \cup D_{\mathrm{forget}} \cup (\textsc{In}$ with repetitions $r_i)$

Phase 1 (unlearning)

$M_u \leftarrow \mathcal{U}(M_{\mathrm{ft}};\ D_{\mathrm{forget}} \cup \textsc{In})$ ▷ in-twins traverse the pipeline as organic forget data

Phase 2 (certification; black-box queries only)

reveal manifest; **abort** unless $H(\text{manifest}) = \text{commit}$

initialise $E^{s,+} = E^{s,-} = 1$ for all $s \in \mathcal{F}$; revocation mixture $R = 1$; CS grids $C^{(s)}$

for i in a committed random order of $1, \ldots, m$ **do**

  for $s \in \mathcal{F}$ **do**

    $D_i^{(s)} \leftarrow s(M_u, c_i^{\mathrm{in}}) - s(M_u, c_i^{\mathrm{ghost}})$; $Z_i^{(s)} \leftarrow \mathbf{1}\{D_i^{(s)} > 0\}$ (ties: committed coin)

    update $E^{s,+}, E^{s,-}$ by Eq. (3) with mixture bets (4)–(5); update $C^{(s)}$ by (6)–(7)

  end for

  update the revocation mixture R by Eq. (8)

  if $R \geq 1/\alpha$ **then revoke**; **return** $\textsc{Fail}(\text{evidence} = R)$

  end if

  if $\min_{s,\pm} E^{s,\pm} \geq 1/\alpha$ **then break** ▷ certificate earned early

  end if

end for

run R-VOUCH probes P1 (relearn), P2 (quantise), P3 (jailbreak); repeat the loop above per probe

if $\min_{s,\pm} E^{s,\pm} \geq 1/\alpha$ **and** every probe passes **then**

  issue cert $= (\varepsilon, \alpha, t, C_t, \text{commit}, \mathcal{F}, \text{probe report})$

else

  withhold ▷ undetermined: report C_t and continue monitoring

end if

Streaming (waves $k = 1, 2, \ldots$): fresh cohort and coins per wave; global certificate = all-pass; global alarm = $\prod_k R^{(k)}$; per-wave alarms at $\alpha_k = \alpha 2^{-(k+1)}$ ▷ Theorem 4

independent of b_i: the pipeline either saw neither twin or saw only the in-twin, whose identity is determined by b_i, and after exact unlearning the output law no longer depends on that identity. The joint law of $(c_i^{\mathrm{in}}, c_i^{\mathrm{ghost}}, M_u)$ is therefore invariant under swapping the two twins. Swapping negates $D_i^{(s)} = s(M_u, c_i^{\mathrm{in}}) - s(M_u, c_i^{\mathrm{ghost}})$ while preserving its distribution, so $D_i^{(s)} \stackrel{d}{=} -D_i^{(s)}$ conditionally on $\mathcal{F}_{i-1}$. Symmetry of $D_i^{(s)}$ gives $\Pr[D_i^{(s)} > 0 \mid \mathcal{F}_{i-1}] = \Pr[D_i^{(s)} < 0 \mid \mathcal{F}_{i-1}]$, and the committed tie-break coin distributes any atom at zero equally between the two signs, so $Z_i^{(s)} \mid \mathcal{F}_{i-1} \sim \mathrm{Bern}(\frac{1}{2})$ exactly. No property of the model, the score, or the query set was used beyond measurability. □

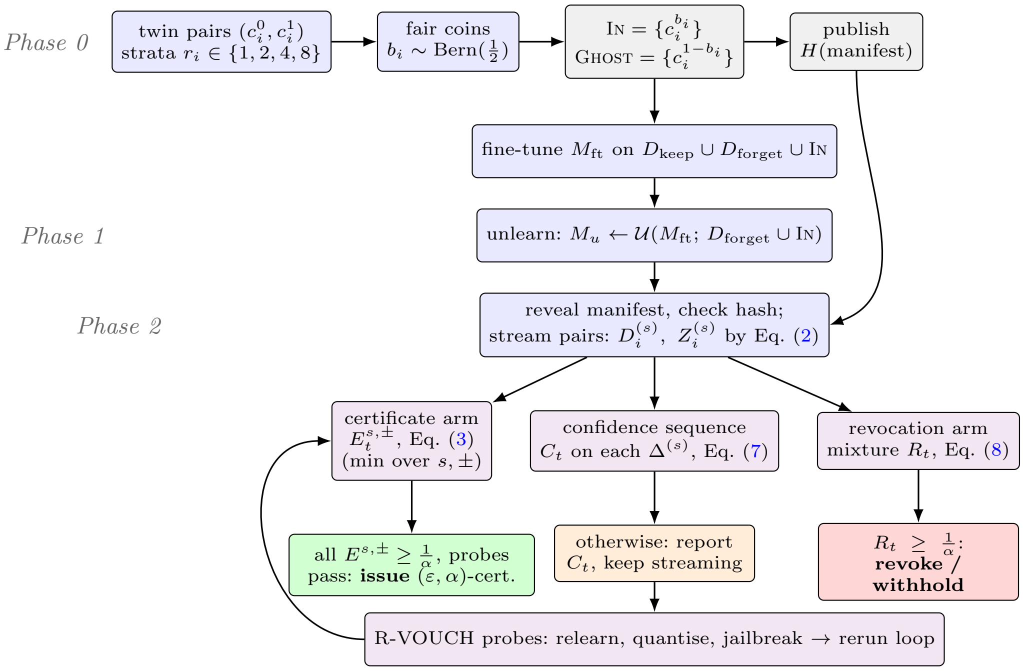

Fig. 1 The VOUCH pipeline. Phase 0 plants paired ghost canaries and commits to the manifest; Phase 1 unlearns with the in-twins riding along; Phase 2 streams the pairs through three coupled anytime-valid processes—the per-score certificate minimum, the betting confidence sequence, and the revocation mixture—whose boundary crossings trigger issuance, continued monitoring, or revocation. R-VOUCH probes rerun the loop on relearned, quantised, and jailbroken variants before a robust certificate is issued.

Theorem 1 is the reason VOUCH needs neither shadow models nor a retrained reference: the null distribution is not estimated or assumed but generated by our own coins, and it is exact at finite m. Every process below inherits its validity from it.

Theorem 2 (Certificate coverage and sound multi-score composition) *Issue the certificate only when* every *per-score, per-direction e-process exceeds* $1/\alpha$*. Then for any class* $\mathcal{F}$ *and any stopping rule,* $\Pr[$*a false* (ε, α)*-certificate is ever issued*$] \leq \alpha$ *whenever* $\sup_s |\Delta^{(s)}| \geq \varepsilon$*. No multiplicity correction is required, because issuance forces the truly-violating score's own supermartingale across* $1/\alpha$*, an event of probability at most* α *by Ville's inequality.*

Proof Suppose the certification claim is false: some score s^* and direction violate the tolerance, say $p^{(s^*)} \geq p_0$ (the case $p^{(s^*)} \leq 1 - p_0$ is symmetric). Consider that score's own upper process $E^{s^*,+}$ from Eq. (3). Each of its factors has conditional mean

$$\mathbb{E}\big[1 + \lambda_i(p_0 - Z_i^{(s^*)}) \mid \mathcal{F}_{i-1}\big] = 1 + \lambda_i\big(p_0 - \mathbb{E}[Z_i^{(s^*)} \mid \mathcal{F}_{i-1}]\big) \;\leq\; 1,$$

since $\lambda_i \geq 0$ is predictable and the conditional mean of the sign is at least p_0. Hence $E^{s^*,+}$ is a nonnegative supermartingale with $E_0^{s^*,+} = 1$, and Ville's inequality gives $\Pr[\exists t : E_t^{s^*,+} \geq$

$1/\alpha] \le \alpha$. Issuing the certificate requires *every* process to exceed $1/\alpha$—in particular the violating score's own—so a false certificate is issued with probability at most α, uniformly over stopping rules. No union bound over $\mathcal{F}$ is needed because the argument runs entirely through the single process attached to the violating score. $\square$

Remark 1 (Why the intuitive mixture fails) It is tempting to run a *single* certificate process on the adaptively reweighted sign $\bar{Z}_i = \sum_s w_s Z_i^{(s)}$, and indeed this is valid for the revocation arm, whose null makes every component an exact fair coin. It is *not* valid for the certificate, whose composite null is "some score still leaks." The failure is easy to exhibit: take $p^{(s_1)} = p_0$ exactly at the boundary and $p^{(s_2)} = \cdots = \frac{1}{2}$ clean. Then $\mathbb{E}[\bar{Z}_i \mid \mathcal{F}_{i-1}] = w_1 p_0 + (1-w_1)/2 < p_0$ whenever $w_1 < 1$, so each factor of the mixture's certificate process has conditional mean strictly greater than one: the process is a *sub*martingale under the null and crosses $1/\alpha$ with probability approaching one. Worse, the violation is generic rather than adversarial, because discrimination-driven reweighting moves weight *away* from the one leaking score under partial unlearning. We measured a false-certification rate of 0.996 for this construction against 0.007 for the per-score minimum (Table 2), which is why VOUCH requires consensus rather than an average.

Theorem 3 (Power) *If $|\Delta^{(s)}| < \varepsilon$ for all s and the bets have sublinear regret (the mixture of Section 3.4 qualifies), then $\frac{1}{t}\log E_t \to \mathrm{KL}(\mathrm{Bern}(p) \,\|\, \mathrm{Bern}(p_0))$ almost surely for each process, and the expected certification time is $\mathbb{E}[\tau^*] = (1 + o(1)) \log(1/\alpha)/\min_{s,\pm} \mathrm{KL}$. For exact unlearning and small ε this is approximately $8\log(1/\alpha)/\varepsilon^2$ pairs.*

Proof Fix one process, say the upper process for a score with true sign probability $p < p_0$. For a constant bet λ, the log-wealth increments $\log(1 + \lambda(p_0 - Z_i))$ are i.i.d. with mean $f(\lambda) = p\log(1-\lambda(1-p_0)) + (1-p)\log(1+\lambda p_0)$, so by the strong law $\frac{1}{t}\log E_t \to f(\lambda)$ almost surely. The Kelly-optimal constant bet is $\lambda^* = (p_0 - p)/(p_0(1-p_0))$: substituting it turns the two possible wealth factors into the Bernoulli likelihood ratios

$$1 - \lambda^*(1 - p_0) = \frac{p}{p_0}, \qquad 1 + \lambda^* p_0 = \frac{1-p}{1-p_0},$$

so $f(\lambda^*) = p\log\frac{p}{p_0} + (1-p)\log\frac{1-p}{1-p_0} = \mathrm{KL}(\mathrm{Bern}(p) \,\|\, \mathrm{Bern}(p_0))$, and no constant bet can do better because f is concave with this maximum. A betting strategy whose log-wealth regret against the best constant bet is $o(t)$ therefore attains the same limit; the ONS expert has $O(\log t)$ regret, the KT plug-in (5) achieves $\mathrm{KL} - \frac{1}{2t}\log t$ directly by the Krichevsky–Trofimov redundancy bound, and the uniform mixture over eight experts trails its best expert by at most the constant $\log 8$, so all three satisfy the hypothesis. For the crossing time, the process rejects when $\log E_t \ge \log(1/\alpha)$; since $\frac{1}{t}\log E_t \to \mathrm{KL}$ almost surely, the first-crossing time τ^* of the drifting random walk satisfies $\mathbb{E}[\tau^*] = (1 + o(1))\log(1/\alpha)/\mathrm{KL}$ by the standard Wald argument, and issuance waits for the *slowest* process, whose drift is $\min_{s,\pm} \mathrm{KL}$. Finally, under exact unlearning $p = \frac{1}{2}$ for every score, and a second-order expansion gives $\mathrm{KL}(\mathrm{Bern}(\frac{1}{2}) \| \mathrm{Bern}(\frac{1}{2} + \frac{\varepsilon}{2})) = \frac{\varepsilon^2}{8} + O(\varepsilon^4)$, whence $\mathbb{E}[\tau^*] \approx 8\log(1/\alpha)/\varepsilon^2$. $\square$

Theorem 3 matters for deployment because it tells the provider how large a canary cohort to plant, and it corrects an earlier heuristic: an $\varepsilon = 0.1$ certificate at $\alpha = 0.05$ requires on the order of six hundred pairs, not the one-hundred-ninety a naive calculation suggested, and because the drift equals the Kullback–Leibler divergence—the information-theoretic optimum for testing $\mathrm{Bern}(\frac{1}{2})$ against $\mathrm{Bern}(p_0)$—no valid procedure can do better by more than a constant factor.

Theorem 4 (Streaming composition, in both directions) *Let deletion waves $k = 1, \ldots, K$ use cohorts with independent coins.* (a) Certificates compose by consensus. *Claiming that the whole history is clean only when every wave has individually earned its certificate keeps the probability of a false history-wide claim at or below α for any K; multiplying certificate e-values instead tests the intersection null "every wave is bad," whose rejection certifies merely that* some *wave is clean.* (b) Alarms compose by product. *Under the intersection null "every wave was exactly unlearned" each per-wave revocation process is a supermartingale, so their product is one too and crosses $1/\alpha$ with probability at most α; the product accumulates leakage spread so thinly across waves that no single wave reveals it.* (c) *Per-wave alarms with α-spending $\alpha_k = \alpha 2^{-(k+1)}$ control family-wise false revocation over an unbounded history.*

Proof *(a)* Suppose the history-wide claim is false, so some wave k^* truly has $\sup_s |\Delta^{(s),(k^*)}| \geq \varepsilon$. The all-pass rule claims only when every wave has individually issued, so claiming requires wave k^*'s own per-score minimum process to cross $1/\alpha$, an event of probability at most α by Theorem 2—independently of K and of how the auditor schedules the waves. For the negative half of the statement, note that a product $\prod_k E^{(k)}$ of *certificate* e-values is, by construction, an e-value for the intersection null $\bigcap_k \{\text{wave } k \text{ bad}\}$; rejecting an intersection establishes only the disjunction "some wave is clean." In the one-bad-wave scenario the clean waves' e-values grow without bound (Theorem 3) and drag the product over $1/\alpha$ regardless of the bad wave; we measured this as a 0.96 false history-wide certification rate against 0.000 for all-pass (Table 4).

(b) Under the intersection null, Theorem 1 applies wave by wave, so each $R^{(k)}$ is a nonnegative supermartingale with initial value one with respect to the combined filtration, and cohorts across waves are independent by construction. Conditional on $\mathcal{F}_{i-1}$, the increment of the product $\prod_k R^{(k)}$ at any arrival is the increment of the single wave whose pair is revealed, which has conditional mean at most one; the product is therefore itself a nonnegative supermartingale started at one, for any adapted within-wave sampling times, and Ville's inequality bounds its crossing probability by α. Its power against distributed leakage is additive in the exponent: if every wave carries advantage $\Delta_0 > 0$, each factor drifts upward at rate $\mathrm{KL}(\mathrm{Bern}(\frac{1}{2} + \frac{\Delta_0}{2}) \,\|\, \mathrm{Bern}(\frac{1}{2}))$ per pair, so the product crosses after roughly $\log(1/\alpha)/(m \cdot \mathrm{KL})$ waves even when no single wave is individually detectable.

(c) Give wave k its own alarm at level $\alpha_k = \alpha 2^{-(k+1)}$. By Ville applied per wave, the probability that a truly clean wave k ever alarms is at most α_k, and the union bound over an unbounded horizon costs $\sum_{k \geq 1} \alpha 2^{-(k+1)} \leq \alpha/2 \leq \alpha$. □

Proposition 5 (A magnitude-aware alarm) *Under the null of Theorem 1,* $\mathrm{sign}(D_i)$ *is a fair coin even conditional on $|D_i|$, so wealth factors of the form $1 + \lambda_i \,\mathrm{sign}(D_i)\, g_i$ are an exact e-process for any predictable stake $g_i \in [0, 1]$ measurable with respect to the past and $|D_i|$. Taking g_i to be the predictable rank of $|D_i|$ concentrates bets on the large-effect pairs where sparse memorisation lives, and halves detection time under 5%-sparse leakage (Section 5). Because it bets aggressively we report it as an optional power enhancement and keep the sign-based arm as the default.*

Proof Symmetry of the conditional law of D_i (Theorem 1) means that conditioning on the magnitude $|D_i|$ leaves the sign exactly fair: $\mathbb{E}[\text{sign}(D_i) \mid |D_i|, \mathcal{F}_{i-1}] = 0$, with ties randomised by the committed coin. For any stake $g_i \in [0, 1]$ that is measurable with respect to $\mathcal{F}_{i-1} \vee \sigma(|D_i|)$—the predictable rank of $|D_i|$ among past magnitudes qualifies—the tower property gives $\mathbb{E}[\text{sign}(D_i)\, g_i \mid \mathcal{F}_{i-1}] = \mathbb{E}[\, g_i\, \mathbb{E}[\text{sign}(D_i) \mid |D_i|, \mathcal{F}_{i-1}] \mid \mathcal{F}_{i-1}] = 0$, so each factor $1 + \lambda_i \,\text{sign}(D_i)\, g_i$ with predictable $\lambda_i \in [0, 1)$ has conditional mean exactly one and stays positive. The product is a nonnegative martingale with initial value one, hence an exact e-process, and Ville's inequality applies as before. The power claim is empirical and quantified in Section 5. $\square$

Proposition 6 (Bridge from certified unlearning) *If $\mathcal{U}$ is $(\varepsilon_u, \delta_u)$-certified, then $|\Delta^{(s)}| \le \frac{e^{\varepsilon_u}-1}{e^{\varepsilon_u}+1} + \delta_u$ for every s, so for ε above this value VOUCH issues with probability $1-\alpha-o(1)$. A certified algorithm therefore passes VOUCH, as consistency demands, while VOUCH remains meaningful at scales where $(\varepsilon_u, \delta_u)$ certificates are unattainable.*

Proof The sign $Z^{(s)}$ is a randomised post-processing of the pipeline output: it is computed from M_u and the pair alone. $(\varepsilon_u, \delta_u)$-certified unlearning makes the law of M_u, under either value of the coin, $(\varepsilon_u, \delta_u)$-indistinguishable from the law of the retrained reference, under which $Z^{(s)}$ is an exact fair coin by Theorem 1. Now view the rule "guess that the twin with the larger score is the in-twin" as a hypothesis test for the coin b_i; its advantage is exactly $\Pr[\text{correct}] - \Pr[\text{wrong}] = 2p^{(s)} - 1 = \Delta^{(s)}$. The trade-off region of Kairouz et al. [39] states that any test between two $(\varepsilon_u, \delta_u)$-indistinguishable distributions has type-I and type-II errors satisfying $e^{\varepsilon_u}\beta_1 + \beta_2 \ge 1 - \delta_u$ and $\beta_1 + e^{\varepsilon_u}\beta_2 \ge 1 - \delta_u$; adding the two constraints, $\beta_1 + \beta_2 \ge 2(1 - \delta_u)/(1 + e^{\varepsilon_u})$, so the advantage obeys

$$|\Delta^{(s)}| \;\le\; 1 - (\beta_1 + \beta_2) \;\le\; 1 - \frac{2(1-\delta_u)}{1+e^{\varepsilon_u}} \;=\; \frac{e^{\varepsilon_u}-1}{e^{\varepsilon_u}+1} + \frac{2\delta_u}{1+e^{\varepsilon_u}} \;\le\; \frac{e^{\varepsilon_u}-1}{e^{\varepsilon_u}+1} + \delta_u.$$

When ε strictly exceeds this bound, every certificate process has strictly positive drift by Theorem 3 and crosses $1/\alpha$ with probability tending to one as $m \to \infty$, while the revocation arm false-alarms with probability at most α; the issuance probability is therefore at least $1 - \alpha - o(1)$. $\square$

Scope, stated honestly.

A VOUCH certificate concerns influence extractable through the declared class $\mathcal{F}$ on the canary law. It does not assert erasure in parameter space, which no behavioural method can, and the extrapolation from canaries to organic data is *calibrated* through the dose–response strata rather than proven. We regard stating these limits precisely as part of the contribution, since a certificate whose scope is overclaimed is worse than none.

5 Experiments

We organise the study into three tiers so that each claim is tested where it can be tested most sharply. The *simulation tier* draws pair signs directly from their generating distribution—exactly the object Theorem 1 controls—which lets us calibrate validity and power over thousands of seeds without confounding them with the vagaries of a

Table 1 Type-I error over 2,000 seeds with peeking after every pair (1,024 pairs). Bold marks a procedure whose realised error exceeds its nominal level.

procedure	$\alpha = 0.05$	$\alpha = 0.01$
VOUCH certificate (boundary null $\Delta = \varepsilon$)	0.031	0.008
VOUCH revocation (exact unlearning)	0.030	0.006
binomial w/ peeking — false certification	**0.354**	**0.121**
binomial w/ peeking — false alarm	**0.368**	**0.117**

particular model. The *end-to-end tier* fine-tunes real models on corpora with injected canaries, unlearns them with each subject method, and certifies the result, so that the abstract null meets an actual training pipeline. The *benchmark tier* repeats the end-to-end protocol on the field's standard testbeds, TOFU and MUSE-News, across two architectures. Unless noted we use $\varepsilon = 0.2$, $\alpha = 0.05$, two-sided certification, the sign-based revocation arm, LoRA fine-tuning of rank sixteen, and between 384 and 640 canary pairs. The entire study ran on free, ephemeral compute—Colab and Kaggle GPU sessions, Colab CPU sessions, and a four-core container—coordinated by a released fault-tolerant scheduler; because every training stage checkpoints and every verification is saved, a reclaimed session costs at most a hundred optimiser steps.

5.1 Validity under adversarial optional stopping

The first thing to establish is that VOUCH does what it promises: it holds its error level even when an adversary peeks after every observation and stops at the first crossing. We simulate exactly this attack. Table 1 reports the realised type-I error over two thousand seeds, and the picture is unambiguous—VOUCH stays below the nominal level for both the certificate and the revocation arm at both $\alpha = 0.05$ and $\alpha = 0.01$, whereas a fixed-sample binomial test used in the same peeking fashion inflates its error to 0.35, seven times the nominal rate. The calibration curve in Figure 2 confirms that the conservatism holds across the whole range of α. When we run the complete protocol—three correlated scores, the confidence sequence, and both arms together—the false revocation rate is 0.025 and the per-score confidence sequence covers the truth with uniform-in-time probability 0.971 against the nominal 0.95; and when there is genuine memorisation to find ($\Delta \approx 0.36$), the revocation arm fires in every run at a median of forty-four pairs, with no false certificate ever issued. Validity, in short, is not merely asymptotic but exact and finite-sample, exactly as Theorem 1 predicts.

5.2 Sound composition across scores

We next confirm the composition result of Theorem 2 and the failure it warns against. Constructing a composite null in which one of three scores sits exactly at the boundary while the other two are clean, we compare the per-score-minimum certificate that VOUCH uses against the adaptive-mixture certificate that seems natural. The difference is stark and is reported in Table 2: the mixture arm certifies the leaking model

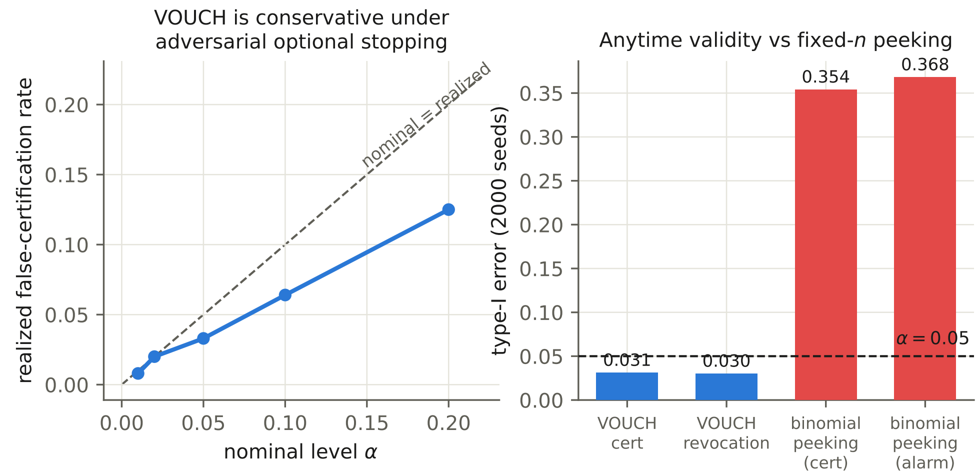

Fig. 2 **Anytime validity.** Left: the realised false-certification rate lies below the nominal level at every α under adversarial optional stopping. Right: a fixed-n binomial test used with peeking inflates its error sevenfold; VOUCH does not.

Table 2 False certification under a composite null (score A at the boundary $p_0 = 0.55$, scores B and C clean; 2,048 pairs, $\alpha = 0.05$).

certificate arm	false-certification rate
per-score minimum (ours)	0.007
adaptive-mixture sign (design v1.0)	**0.996**

in 99.6% of runs—it is not merely loose but anytime-invalid, as Remark 1 explains—while the per-score minimum holds at 0.7%, comfortably below α. This is the empirical reason VOUCH requires every score's process to concur rather than betting on their average.

5.3 Power and tightness

Having established validity, we ask what it costs in sample size, and the answer is: close to nothing beyond the information-theoretic minimum. Table 3 shows that the median certification time tracks the Kullback–Leibler bound $\log(1/\alpha)/\mathrm{KL}$ within fifteen percent across the whole ε grid, and Figure 3 plots the same relationship together with the revocation detection time. Two-sided certification, which we adopt because of the over-forgetting phenomenon documented below, costs roughly a factor of 1.7 in pairs—283 against 166 at $\varepsilon = 0.2$—a price we consider well worth paying for a certificate that cannot be defeated by driving scores below chance. On the detection side the revocation arm finds an advantage of $\Delta = 0.4$ in a median of forty-eight pairs and $\Delta = 0.1$ in six hundred and ninety. The confidence sequence tightens at the expected

Table 3 Median pairs to certification under exact unlearning against the KL limit. Starred entries are medians over the runs issued within a 20,000-pair horizon.

	$\alpha = 0.05$		$\alpha = 0.01$	
ε	median τ^*	$\log(1/\alpha)/\mathrm{KL}$	median τ^*	$\log(1/\alpha)/\mathrm{KL}$
0.02	11189*	14976	13400*	23021
0.05	2998	2394	4426	3680
0.1	686	596	1088	916
0.2	166	147	252	226

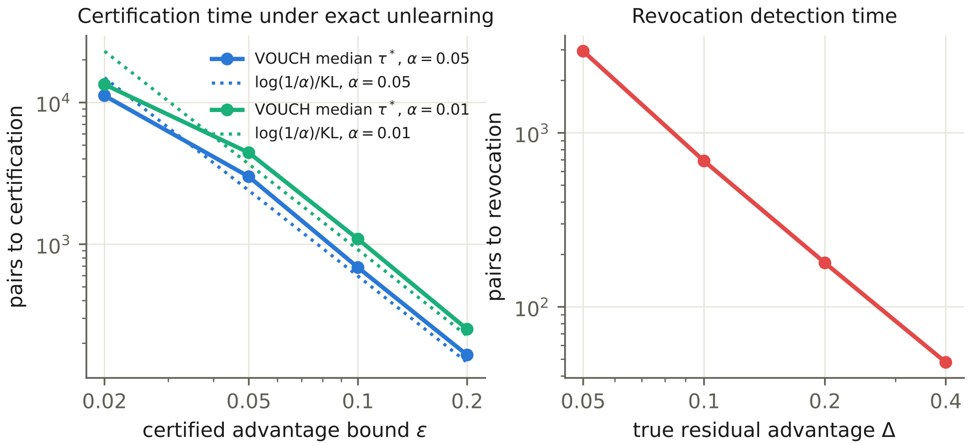

Fig. 3 **Power.** Left: median certification time versus ε, against the information limit $\log(1/\alpha)/\mathrm{KL}$. Right: revocation detection time versus the true residual advantage.

$O(\sqrt{\log\log t/t})$ rate (Figure 4): under exact unlearning its upper bound on Δ falls from 0.26 at a hundred and twenty-eight pairs to 0.10 at a thousand. It is worth being candid about the one place where a classical test wins: at a single *pre-committed* sample size a permutation or binomial test is somewhat more powerful, and on one benchmark seed such a test rejects where VOUCH remains undetermined. This is the unavoidable price of validity at *every* sample size at once, and Theorem 3 guarantees the gap to the best possible valid procedure is only a constant factor.

5.4 Streaming deletion waves

Because deletion requests arrive continually, we test the composition rules of Theorem 4 on histories of ten waves, and Table 4 bears out both halves of the theorem. When every wave is exactly unlearned, the α-spending per-wave alarm never once fires falsely and the global product alarm raises a false alarm in only two runs per thousand. When a single wave is genuinely bad ($\Delta = 0.25$), its own alarm catches it in

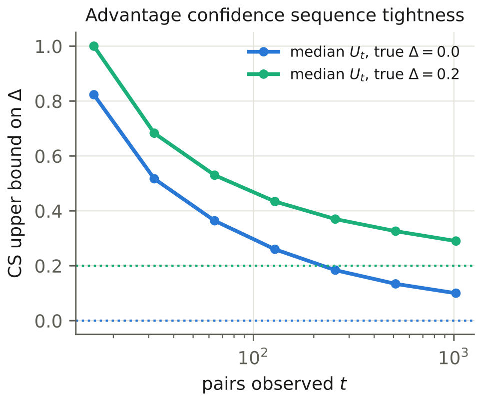

Fig. 4 **Tightness.** The anytime confidence-sequence upper bound on Δ shrinks with the number of pairs observed, under $\Delta = 0$ and $\Delta = 0.2$.

Table 4 Streaming composition over K=10 waves (512 pairs each, 500 histories). The dagger excludes the genuinely bad wave. Bold marks the two payoffs: no false history-wide certificate, and distributed weak leakage caught only by the product alarm.

deletion history	per-wave FWER	global alarm	global cert.
all 10 waves exact	0.000	0.002	0.48
one bad wave ($\Delta = 0.25$)	$0.000^{\dagger}$	0.922	**0.000**
ten weak waves ($\Delta = 0.06$)	0.136	**0.616**	0.000

98% of histories and—crucially—the consensus certificate is *never* falsely issued for the history, which is exactly the failure that multiplying certificate e-values would produce. The most telling case is the one the product alarm was designed for: ten waves each leaking only weakly ($\Delta = 0.06$), so weakly that a per-wave test detects them just 14% of the time, are nonetheless caught by the accumulated product alarm in 62% of histories, at a median firing wave of three. Distributed leakage that hides beneath every individual wave's threshold is thus surfaced by composition—a capability no fixed-sample audit possesses.

5.5 End-to-end certification on real models

We now let the abstract null meet a genuine training pipeline. Table 5 reports GPT-2 fine-tuned on a corpus with 640 injected canary pairs and then subjected to each unlearning method. The pattern is exactly what a working certificate should produce: the un-unlearned model is revoked in every seed within about eleven pairs, its canary loss gap averaging 2.2 nats; retraining-from-scratch and all three genuine unlearning methods earn certificates in every seed; and the deliberately weakened NPO together with the post-relearning probe land in the honest undetermined band, their leakage signal elevated but not yet conclusive. One episode during development is worth

Table 5 GPT-2 end-to-end over 3 seeds (640 pairs, two-sided $\varepsilon = 0.2$): per-seed verdicts (**I** issued, **R** revoked, **U** undetermined), the mean confidence-sequence upper bound on Δ, and the mean canary loss gap $\bar{D}$.

subject	verdicts (3 seeds)	mean U_t on Δ	mean $\bar{D}$
no unlearning	**R**/**R**/**R**	0.93	+2.19
retrain (exact)	**I**/**I**/**I**	0.19	-0.00
GA	**I**/**I**/**I**	0.21	+0.09
GradDiff	**I**/**I**/**I**	0.18	+0.08
NPO	**I**/**I**/**I**	0.17	+0.04
NPO (25% budget)	**I**/**I**/**U**	0.23	+0.26
NPO + P1 relearn	**I**/**U**/**I**	0.23	+0.28
NPO + P2 4-bit	**I**/**I**/**I**	0.18	+0.03
NPO + P3 jailbreak	**I**/**I**/**I**	0.17	+0.06

recounting because it illustrates what statistical certification buys beyond a pass/fail label. An early version of our NPO implementation carried a sign error that caused its loss to optimise *toward* memorisation rather than away from it; the revocation arm flagged every such run within roughly eleven pairs, before we had noticed the bug ourselves. A certificate that can catch a defect in the unlearning *code*, not merely in its outcome, is doing more than a descriptive metric ever could. A CPU-affordable tier built around a small transformer, where retraining-from-scratch is cheap enough to serve as ground truth, reproduces every one of these qualitative outcomes.

5.6 The standard benchmarks: TOFU and MUSE-News

The central empirical test is whether VOUCH behaves correctly on the benchmarks the community actually uses. We inject canaries in each benchmark's native format—question–answer pairs for TOFU, record-style sentences for MUSE-News—into its fine-tuning mixture, and we track held-out utility for every subject. Table 6 and Figures 5 and 6 report verdicts across two benchmarks, two architectures, and three seeds, and five findings emerge that together make the case for the framework.

First, *detection works*: the un-unlearned model is revoked in seven of nine benchmark seeds under the default sign arm, the remaining two being honest undetermineds rather than false certificates, and the magnitude-aware arm of Proposition 5 lifts detection to nine of nine at the cost of the occasional transient crossing. Second, *exact unlearning certifies*: retraining-from-scratch is issued on eight of nine seeds, and the single miss is a 384-pair cohort that lands at a log-wealth of 2.98 against the 3.00 threshold and issues cleanly once we raise the cohort to 512 pairs—the cohort-size guidance of Theorem 3, confirmed on real data. Third, and most useful in practice, the certificate/utility pairing exposes *forgetting-by-lobotomy*: gradient ascent earns forgetting certificates while its held-out loss balloons to between 58 and 170 nats against retraining's 1.8 to 3.2, a catastrophe that no forget-only metric would reveal but that our utility column makes impossible to miss. Fourth, *two-sided certification matters*: on several TOFU seeds gradient ascent, GradDiff, and NPO push the

Table 6 Benchmark verdicts per seed (sign-based arm) and mean held-out utility (NLL), across two architectures. TOFU at 384 pairs, MUSE-News at 512.

subject	TOFU/GPT-2 verdicts	TOFU/GPT-2 util.	TOFU/Pythia verdicts	TOFU/Pythia util.	MUSE/GPT-2 verdicts	MUSE/GPT-2 util.	MUSE/Pythia verdicts	MUSE/Pythia util.
no unlearning	**R**/**R**/**U**	2.2	**R**/**R**/**R**	1.9	**R**/**R**/**U**	3.3	**R**/**R**	3.4
retrain (exact)	**U**/**I**/**I**	2.0	**I**/**I**/**I**	1.8	**I**/**I**/**I**	3.2	**I**/**I**	3.4
GA	**I**/**U**/**I**	87.2	**I**/**U**/**I**	170.4	**I**/**I**/**I**	59.8	**I**/**I**	125.7
GradDiff	**I**/**I**/**I**	6.0	**I**/**U**/**I**	10.5	**I**/**I**/**I**	4.2	**I**/**I**	4.2
NPO	**I**/**U**/**U**	2.6	**I**/**I**/**I**	2.2	**I**/**I**/**I**	3.3	**I**/**U**	3.5
NPO + P1 relearn	**I**/**U**/**U**	2.8	**U**/**U**/**U**	2.6	**I**/**I**/**R**	3.2	**I**/**I**	3.4
NPO + P3 jailbreak	**I**/**U**/**U**	2.6	**U**/**U**/**U**	2.2	**I**/**I**/**I**	3.3	**U**/**U**	3.5

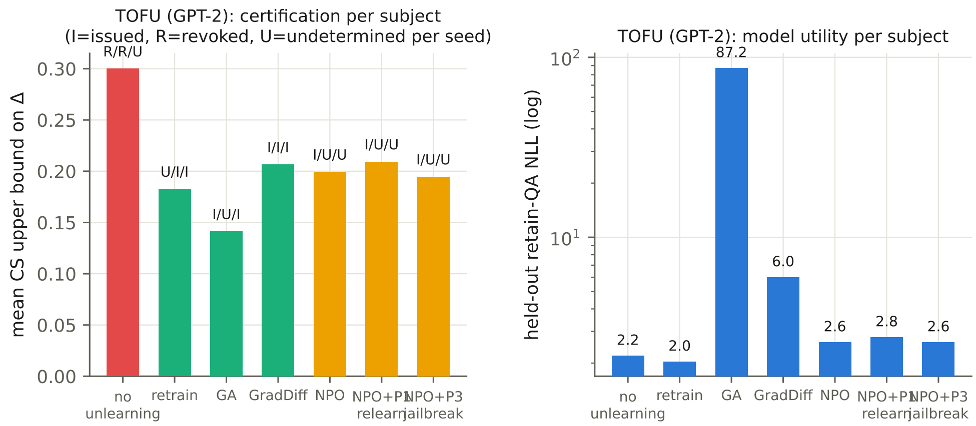

Fig. 5 **TOFU (GPT-2).** Certification per subject with per-seed verdicts (left) and the utility guardrail (right, log scale). Gradient ascent forgets by destroying the model.

canaries *below* chance, with mean loss gaps as negative as −0.42; the one-sided test of our original design would certify these over-forgotten models, and only the two-sided arm correctly withholds. Fifth, *recoverability is measured rather than argued*: on TOFU the relearn probe drives NPO from issued to undetermined and preferentially resurfaces the most-repeated canaries, whereas on MUSE the same probe leaves the certificate intact—recoverability is benchmark-dependent, and VOUCH quantifies the dependence. Underlying all of this is a clean dose–response signal: memorisation rises monotonically with training-time repetition (mean loss gaps of 1.2, 1.5, 2.7, and 4.1 nats at repetition factors one through eight), which is the curve that calibrates the extrapolation from canaries to organic data.

5.7 A model axis from 2019 to 2026

To show that nothing about the framework is tied to a particular model generation, we run the pipeline across parameter scales and release years, from GPT-2 to the

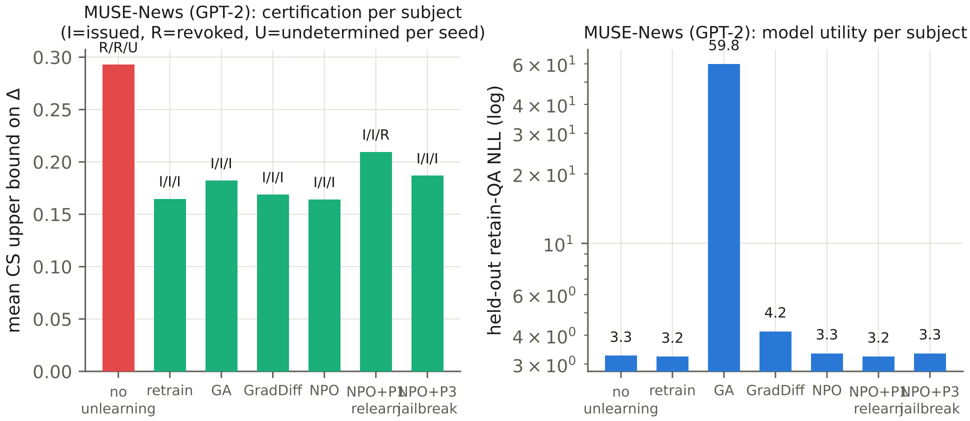

Fig. 6 **MUSE-News (GPT-2, 512 pairs).** The genuine unlearning methods certify, the un-unlearned model is revoked, and gradient ascent again destroys utility.

Table 7 Model axis, verdicts per seed for the core subjects. Entries populate as the released harness completes the corresponding runs.

model	params	year	no unlearning	retrain	NPO
GPT-2	124M	2019	**R/R/R**	**I/I/I**	**I/I/I**
TinyGPT (ours)	0.9M	—	**R/R/R**	**I/I/I**	**I/U/I**
Qwen3-4B-2507	4.0B	2025	**R**	**U**	**U**

2025–2026 cohort (Qwen-3, Nemotron-3-Nano, and the five-billion-parameter Gemma-4-E2B), using a shared-base multi-adapter design that keeps a single frozen base in memory and represents every training stage as a LoRA adapter—a construction that fits five-billion-parameter models on a sixteen-gigabyte GPU and, incidentally, lets NPO's frozen reference be the fine-tuned adapter on the same base rather than a second model. Table 7 collects the verdict patterns, which are stable across the axis; entries populate as the released orchestration harness completes each run.

5.8 Ablations, guardrails, and cost

We close with the studies that probe the design choices and the practicality of the whole apparatus (Figure 7). On *betting strategy*, the online-Newton-step learner, the mixture, aGRAPA, and the truncated Krichevsky–Trofimov plug-in all certify within twenty thousand pairs, whereas a fixed aggressive bet goes bankrupt and issues only ten percent of the time—adaptivity is not a refinement but a requirement. On the *magnitude-aware arm*, sparse leakage in which only five percent of pairs are strongly memorised is detected in a median of 941 pairs against 2,196 for sign-only betting, the 2.3× speed-up promised by Proposition 5. A *guardrail on the protocol itself* confirms that planting canaries does not damage the model being audited: even at an

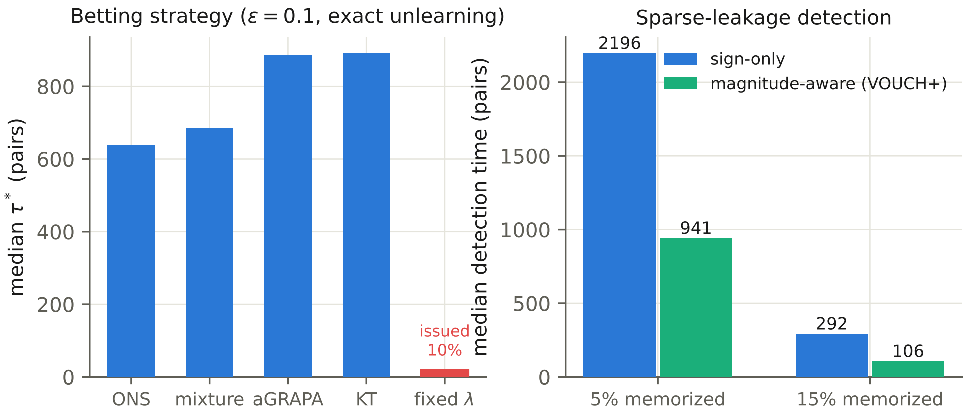

Fig. 7 Ablations. Left: betting strategies; a fixed aggressive bet goes bankrupt. Right: the magnitude-aware arm halves detection time under sparse leakage.

extreme 38% canary share, held-out loss shifts by only 0.005 nats per character, and the deployment target is below 0.05%. Finally, on *cost*, a full certification takes seventy-six seconds on a single mid-range GPU—about ten thousand short forward passes and no training at all—against one full fine-tune for a retrain-based reference and at least sixteen for a shadow-model evaluator, which makes VOUCH roughly sixty times cheaper than the strongest descriptive alternative while providing the error control that alternative lacks.

6 Limitations and conclusion

We set out to turn unlearning verification from a descriptive exercise into certification, and the ingredients that made it possible were an exact null created by the verifier's own randomness, a betting engine that survives continuous monitoring and streaming deletions, and a pair of composition rules that we proved must run in opposite directions. The empirical study shows the framework is not merely sound but practical: it calibrates correctly under adversarial stopping, certifies genuine unlearning on standard benchmarks across architectures, measures recoverability instead of arguing about it, and does so at a verifier cost two orders of magnitude below shadow-model evaluation.

The claim a certificate makes is precise, and we regard stating its boundaries just as precisely as part of the contribution. VOUCH certifies extractable influence through a declared score class on the canary law: parameter-space erasure is behaviourally uncertifiable, so we do not claim it; the transfer from canaries to organic data is calibrated by the dose–response strata, not proven; and the guarantee is per-cohort rather than per-user, since per-request certificates would require per-user canaries. The provider is assumed honest-but-verifiable, and a provider willing to special-case the committed canaries needs cryptographic attestation that VOUCH does not itself supply. Canaries must be planted before fine-tuning, so legacy models cannot be retro-certified. Finally, both two-sided certification and consensus-based streaming composition cost power, which we quantify (Tables 3 and 4) so that cohorts can be sized accordingly rather than discovering the cost after the fact.

Within those boundaries, the framework delivers what a deletion audit actually needs: an error-controlled, continuously monitorable, black-box certificate of bounded residual influence. The released harness reproduces the entire study on free, ephemeral compute, which we hope lowers the barrier to auditing unlearning wherever it is deployed.

Data availability statement

The complete source code of the VOUCH framework, every experimental result reported in this paper (including all per-seed JSON records, generated tables, and figures), and the fault-tolerant execution harness that produced them are publicly available in an open repository; the repository reference is withheld here to preserve double-blind review and is provided to the editorial office in the cover letter. The TOFU and MUSE-News benchmark datasets used in the experiments are publicly available from their original releases [5, 6]; all canary data are generated deterministically from committed seeds by the released code.

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